Review


```
"Summary": "The paper proposes a novel layer-wise learning rate strategy to
accelerate and enhance the grokking phenomenon in Transformer models. The
approach involves assigning different learning rates to the embedding
layers, lower Transformer layers, and higher Transformer layers. The method
is empirically validated on algorithmic tasks such as modular arithmetic
and permutations, showing significant improvements in convergence speed and
final performance.",
"Strengths": [
    "The paper addresses an important problem in deep learning: the
grokking phenomenon.",
    "The proposed layer-wise learning rate strategy is novel and shows
significant improvements in experimental results.",
    "Experiments demonstrate substantial improvements in both convergence
speed and final performance."
],
"Weaknesses": [
    "The paper lacks detailed methodological clarity, particularly
regarding the exact implementation of the layer-wise learning rates and
hyperparameter tuning.",
    "The theoretical explanation for why layer-wise learning rates work is
insufficient.",
    "The scope of tasks is limited to algorithmic ones, making it unclear
how well the findings generalize to other domains.",
    "The choice of learning rates seems arbitrary and lacks
justification.",
    "More comprehensive ablation studies and comparisons with other related
methods would strengthen the paper.",
    "Certain sections, such as the experimental setup and ablation studies,
could be more detailed and clearer."
],
"Originality": 3,
"Quality": 2,
"Clarity": 3,
"Significance": 3,
"Questions": [
    "Can the authors provide more detailed explanations of the
hyperparameter tuning process and the exact implementation of the layer-
wise learning rates?",
    "How do the authors ensure that the proposed method generalizes to
tasks beyond the algorithmic ones tested in the paper?",
    "Can the authors compare their approach with other related methods in
more detail?",
    "Can you provide more theoretical insights into why layer-wise learning
rates specifically facilitate grokking?",
    "How were the specific learning rates chosen for embedding, lower, and
higher layers?",
    "Can you discuss the potential for overfitting and how it was
mitigated?",
    "Have you tested the robustness of your method across different
datasets and larger model sizes?",
    "What is the impact of different learning rate configurations on the
results?",
    "Can the authors discuss potential strategies for mitigating the need
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for careful tuning of learning rates to avoid instability?"
],
"Limitations": [
    "The methodology lacks detailed clarity, and the authors do not provide
sufficient information on the hyperparameter tuning process.",
    "The scope of tasks is limited to algorithmic ones, and the
generalizability of the findings is unclear.",
    "The paper requires more theoretical backing for the proposed method.",
    "The choice of specific learning rates and potential overfitting issues
need to be addressed in more detail.",
    "The scalability of the approach to larger models and more complex
tasks is not thoroughly addressed.",
    "Ethical concerns related to the potential misuse of accelerated
learning techniques are not addressed."
],
"Ethical Concerns": false,
"Soundness": 2,
"Presentation": 2,
"Contribution": 3,
"Overall": 4,
"Confidence": 4,
"Decision": "Reject"
```

D.9. Grokking Through Compression: Unveiling Sudden Generalization via Minimal Description Length

This idea was proposed in the 22nd iteration of a Sonnet 3.5 run.

Idea

```
"Name": "mdl_grokking_correlation",
"Title": "Minimal Description Length and Grokking: An Information-Theoretic
Perspective on Sudden Generalization",
"Experiment": "Implement a function estimate_mdl(model) using weight
pruning to approximate the model's description length. Prune weights below
a threshold and count remaining non-zero weights. Modify the training loop
to compute MDL every 500 steps. Run experiments on ModDivisionDataset and
PermutationGroup, including a baseline without MDL tracking. Plot MDL
estimates alongside validation accuracy. Define the 'MDL transition point'
as the step with the steepest decrease in MDL. Compare this point with the
grokking point (95% validation accuracy). Analyze the correlation between
MDL reduction and improvement in validation accuracy. Compare MDL evolution
between grokking and non-grokking (baseline) scenarios.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 9,
"novel": true
```

Link to code: https://github.com/SakanaAI/AI-Scientist/tree/main/example_papers/mdl_grokking_correlation.